

⇒ CIVIC FIX

⇒ Smart Public Problem Reporting
& Resolution System
⇒ Web-based application.

Introduction:

Civic Fix is a web based system designed to provide citizens with a centralized platform for reporting & tracking public problems like potholes, broken streetlights etc. Citizens can submit complaints with its description, category, location & photographic evidence. Administrators can review complaints, assign them to relative departments or workers, and monitors their progress until resolution.

Users:

Citizen , Administrator ,
Department / Worker.

Stakeholders:

- Citizens : Report & Track.
- Administrators: Manage the process.
- Municipal/Public Service Departments:
Receive & resolve problems.
- Field Workers: Perform & update.
- System Developers: Develop, Maintain
& improve the platform.
- Government/ Local authorities:
Potential future users of
reports & statistics.

Functional Requirements:

User Registration & login: Allows
user to create accounts & login.

Complaint Submission: Citizen able
to submit complaints.

Complaint Tracking: System shall
generate a unique complaint id.

Complaint Verification: Administrators
shall be able to review & verify.

Priority Assignment: Admins assign
priorities low, Medium & High.

Department Assignments: Admins

assign complaints to its department
Status Updates: Authorized Users
able to update status.

Notifications: System notify citizens.

Feedback: Citizens provide
feedback.

Dashboard & Reports: Admins shall
be able to view stats, by category,
priority, department & status.

Non-Functional Requirements:

Security: Protect all data

Performance: Respond in time

Usability: Simple & Understandable

Reliability: Accurate & Reliable

Scalability: Support the traffic of users.

Maintainability: Well organized code

Availability: Should remain accessible

Data Integrity: Prevent unauthorized
or accidental modification.

Risks:

Fake Complaints, Duplicate Complaints,
Incorrect Location, Unauthorized access,

Malicious file uploads, Data loss,
Server downtime, AI misclassification,
Scope becoming too large.

Quality Attributes:

Security: Protect all data

Usability: Able to report without difficulty.

Reliability: History remain accurate.

Performance: Should handle requests.

Data Integrity: Complaints are
accessed by authorized users.

Where AI can be used:

- Automatic Complaint Classification
- Duplicate Complaint detection
- Priority Recommendation
- Complaint Summarization
- Image Analysis

Where AI can be harmful:

- Wrong Classification
- Wrong Priority
- False or Manipulated Image
- Privacy Issues

- AI Bias
- Overdependence on AI

Conclusion:

Civic Fin is designed to provide a structured solution for reporting & Tracking public problems.

The project focuses on important Software Engineering problems & principles including Requirement Analysis, Security, Reliability, Maintainability & Agile development.

AI can enhance Civic Fin through complaint classification, duplicate detection, priority recommendations, summarization & image analysis. However, AI should remain an assisting component rather than final decision maker because incorrect predictions, Privacy issues & bias can affect the system negatively.